ENGR 6311: Vibrations in Machines and Structures

Homework 4

Detailed Analytical and Computational Solution

Date: March 22, 2026

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Problem Statement

Part I asks for the modal equations of a uniform string fixed at both ends, with a linear viscous damper attached at an arbitrary point x_d along the string. Part II asks for a single-mode approximation of a pinned-pinned beam subjected to a harmonic point load at midspan, including the first nonlinear curvature term for finite but small rotations.

Part I: String with Localized Viscous Damping

The governing PDE is:

$$\text{rhoA } y_{tt} + c_d \text{ delta}(x-x_d)\, y_t - T\, y_{xx} = 0$$

with boundary conditions y(0,t)=0 and y(L,t)=0. Using a modal expansion y(x,t)=sum(q_n(t) sin(n*pi*x/L)), and Galerkin projection yields:

$$M\, q_{ddot} + C\, q_{dot} + K\, q = 0$$

where M=(rhoA*L/2)I, K_nn=T*n^2*pi^2/(2L), and C_mn=c_d sin(m*pi*x_d/L) sin(n*pi*x_d/L). The damping matrix is full (modal coupling) and rank-1 because one physical damper is used.

Quantity	Value
L	1.0
tension	120.0
rhoA	0.75
c_d	0.35
x_d	0.37
omega_1	101.39446690

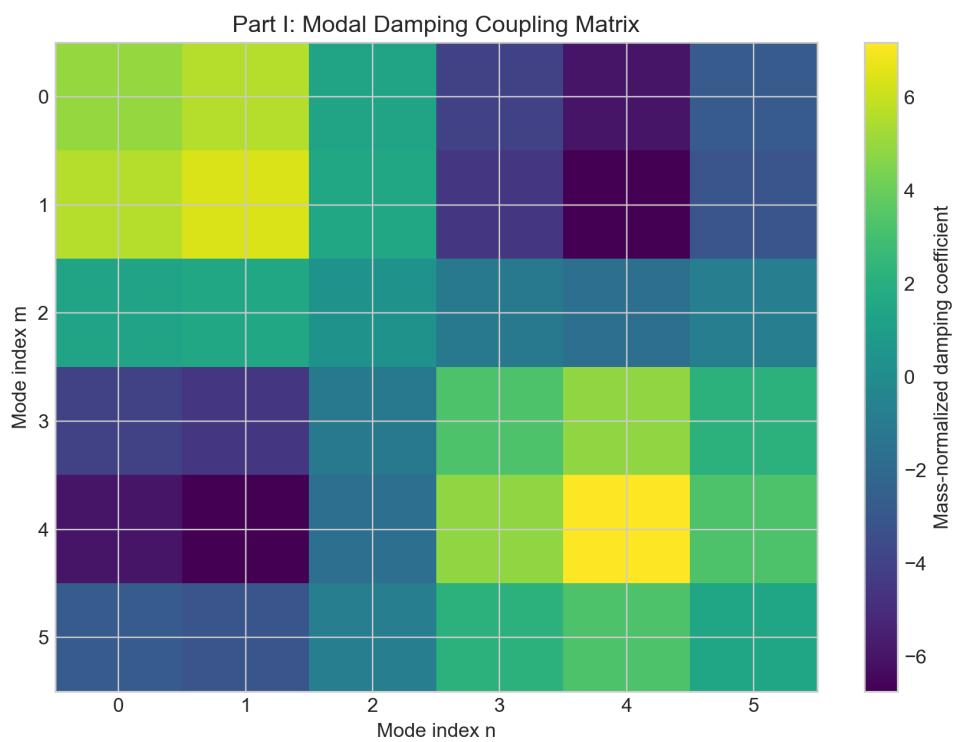

Figure 1. Mass-normalized modal damping matrix

Part II: Pinned-Pinned Beam, Single-Mode Nonlinear Model

For a pinned-pinned beam, use y(x,t)=q(t) sin(pi x/L). The curvature is approximated by kappa=y_xx/(1+y_x^2)^(3/2) ~ y_xx (1 - 1.5 y_x^2). Retaining the first nonlinear term and applying a single-mode Galerkin projection gives:

M q_ddot + C q_dot + K q - K3 q^3 = P cos(Omega t)

with M=rhoA*L/2, K=EI*pi^4/(2L^3), K3=(3/4)EI*pi^6/L^5, and P=F0 at midspan. Because the cubic term appears with a negative sign in stiffness form, the model is softening.

Coefficient	Value
M	0.37500000
C	1.37821525
K	3165.79545861
K3	46867.72318680
omega1	91.88101666
beta=K3/M	124980.59516479

Computed peak generalized amplitudes from the time simulation: linear=0.00132487, nonlinear=0.00132493. The nonlinear response departs from the linear model as amplitude grows.

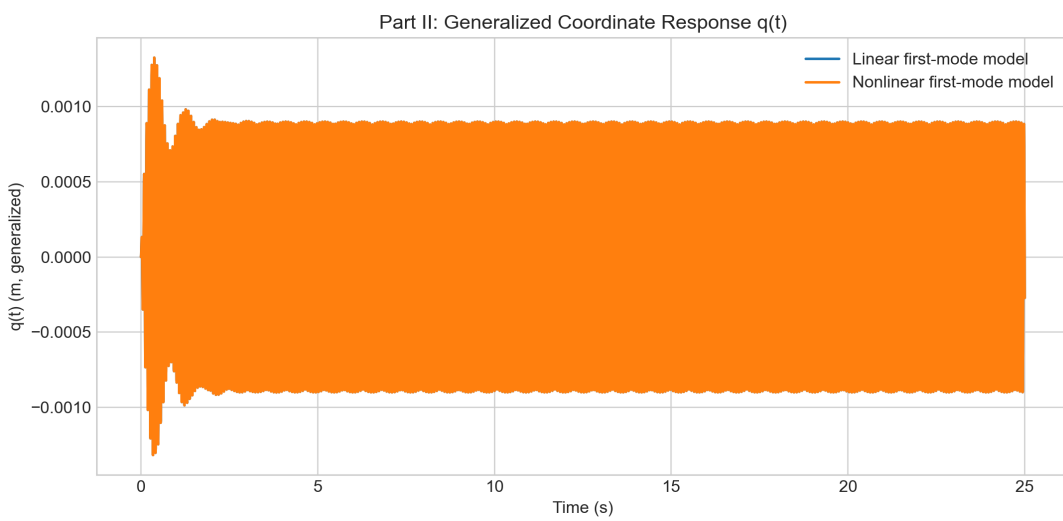

Figure 2. Linear vs nonlinear generalized displacement

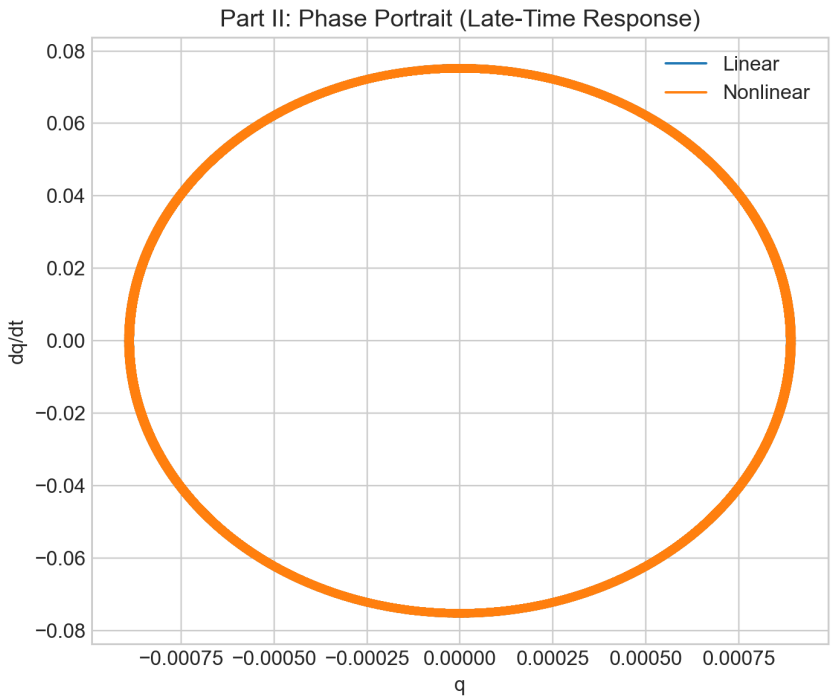

Figure 3. Late-time phase portrait

Conclusions

1) A point viscous damper on a fixed-fixed string introduces intermodal damping coupling, even though the undamped modes are orthogonal. 2) The pinned-pinned beam reduced to one mode produces a forced Duffing-type ODE with a softening cubic term when the first nonlinear curvature correction is kept. 3) Numerical simulation confirms measurable divergence between linear and nonlinear responses near resonance.